# Taller: Control Robusto y Estocástico

## Tema: Modelando las incertidumbres


## Dr. Julio A. García Rodríguez


Use the `ureal` uncertain element to represent real numbers whose values are uncertain when modeling dynamic systems with uncertainty. An uncertain real parameter has a nominal value, stored in the `NominalValue` property, and an uncertainty, which is the potential deviation from the nominal value. `ureal` stores this deviation equivalently in three different properties: `PlusMinus, Range and Percentage.`

```
clc, clear all, close all

delta1 = ureal('delta1',2);
delta2 = ureal('delta2',2,'Percentage',[-10 20]);

get(delta1)
get(delta2)
```

You can create an uncertain real parameter whose value can vary from 14 to 19, with a nominal value of 15.5.

```
p1 = ureal('p1',15.5,'Range',[14,19])
get(p1)
```

Create an uncertain real parameter with a nominal value of 24, whose value can increase or decrease by 15%.

```
k1 = ureal('k1',24,'Percentage',15)
get(k1)
```

Use the following to create a `ureal` parameter and an uncertain dynamic system model.

```
prm_1 = ureal('w0',10);
sys_1 = tf(prm_1,[1 prm_1])
sys_1.NominalValue
```

You could make a simple plot of your parameters

```
pmin = prm_1.Range(1);
pmax = prm_1.Range(2);
pnom = prm_1.NominalValue;

figure
```

```
plot([pmin pmax],[1 1],'k-','LineWidth',2)
hold on
plot(pnom,1,'bo','MarkerSize',12,'LineWidth',3)

xlim([pmin-1 pmax+1])
ylim([0.5 1.5])
yticks([])
grid on

xlabel('Value of Parameter 1')
title('Uncertainty interval')
legend('Interval','Nominal')
```

Plot multiple uncertain parameters

```
delta1 = ureal('delta1',2);
delta2 = ureal('delta2',2,'Percentage',[-10 20]);
p1 = ureal('p1',15.5,'Range',[14 19]);
k1 = ureal('k1',24,'Percentage',15);

parameters = {delta1,delta2,p1,k1};
names = {'delta1','delta2','p1','k1'};

figure
hold on, grid on

for i = 1:length(parameters)
    parameter = parameters{i};
    pmin = parameter.Range(1);
    pmax = parameter.Range(2);
    pnom = parameter.NominalValue;
    plot([pmin pmax],[i i],'k-','LineWidth',2)
    plot(pnom,i,'bo','MarkerSize',8,'LineWidth',3)
end

yticks(1:length(parameters))
yticklabels(names)

xlabel('Parameter Value')
title('Intervals')
legend('Interval','Nominal')
```

## Exercise 1.

Create a model of a second-order system with natural frequency w0 = 10 ±3 rad/s and a damping ratio that can vary from 0.5 to 0.8, with a nominal value of $\zeta$ = 0.6. First, represent the natural frequency and damping ratio values as uncertain real parameters.

```
% Parameters
w0 = ureal('w0',10,'PlusMinus',[-3 3]);
zeta = ureal('zeta',0.6,'Range',[0.5 0.8]);
```

```
% System
sys1 = tf(1,[1/w0^2 2*zeta/w0 1])
figure
step(sys1, 'm--', sys1.NominalValue,'r-')
legend('Uncertain', 'Nominal')
```

## Exercise 2.

Create a state-space model with matrices having uncertain elements. **Tip:** you could first use uncertain parameters such as `ureal` and `ucomplex` and then use these elements to specify the state-space matrices of the system.

```
p = ureal('p',1);

A = [0 3*p; -p p^2];
B = [0; p];
C = ones(2);
D = zeros(2,1);
usys = ss(A,B,C,D)
usys.Nominal
```

## Exercise 3.

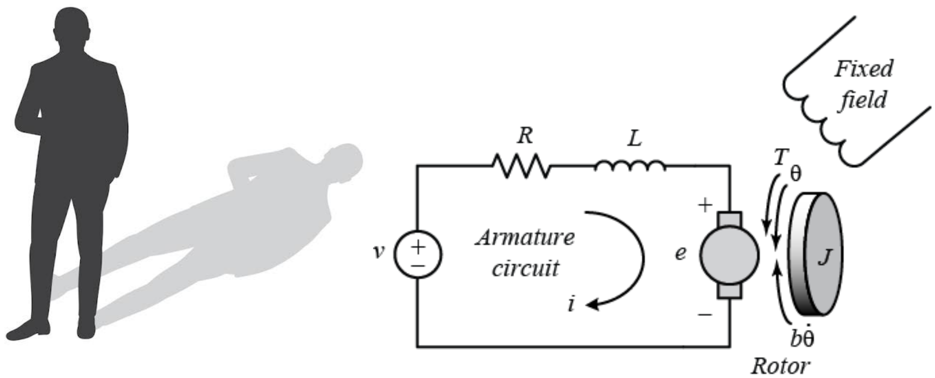
Consider an RC circuit with transfer function $G(s) = \dfrac{1}{RCs + 1}$ with $C = 100\mu F$ and $R = 1K\Omega$. Find the step response when the Resistor and Capacitor tolerance varies 5%, 10% and 15%.

```
R = ureal('R',1000,'Percentage',15);
C = 100e-6;
%C = ureal('C',100e-6,'Percentage',10);
G = tf(1,[R*C 1]);

figure
step(G,G.NominalValue)
grid on
legend('Uncertain','Nominal')
title('Step Response')
```

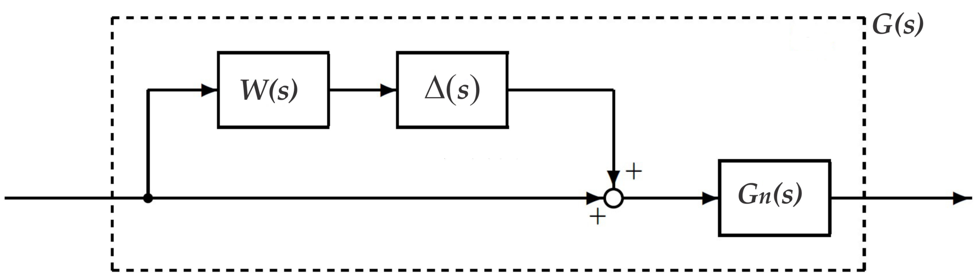
The `ultidyn` objects represent uncertain linear time-invariant dynamics whose only known attributes are bounds on their frequency response. Combine `ultidyn` objects with other dynamic system models and uncertain elements to model systems with uncertain dynamics, represented by `uss` models.

Shadow-Men Analogy




Suppose you want to make the transfer function identification of a dc motor and you find the nominal model $G_n(s) = \dfrac{5}{0.2s + 1}$.

This model describes correctly the model behavior in low-frequencies. However, the motor has elasticity in the axis, mechanical vibrations, structural resonances and other unknown dynamics. Only we know is that its effect is limited. In this case the real plant can be represented as $G(s) = G_n(s)(1 + W(s)\Delta(s))$




where *W(s)* is the weighting function and $\Delta(s)$ is the unknown dynamics with $||\Delta_\infty|| \leq 1$.

## Exercise 4.

Create an uncertain dynamic system model representing an uncertainty with gain of 0.1 at low frequencies and gain of 10 at high frequencies.

```
H = ultidyn("H",1)              %Creation of a SISO ultidyn block with gain less
than 1 at all frequencies.
W = tf([1 0.1],[0.1 1])         %Creation a weighting function with gain starting a
0.1 and increasing to 10.
bodemag(W,{1e-2,1e2}), grid on
bodemag(W,'ro',W*H,'m-'), grid on, ylim([-60 30])
```

You can combine `Delta` with other models to introduce the dynamic uncertainty to your system. For example to model a system as a state-space

```
H2 = ultidyn("H",2)
Delta = W*H2
A = [-5 10;-10 -5];
B = [1 0;0 1];
C = [1 10;-10 1];
D = 0;
Gnom = ss(A,B,C,D);

Gadd = Gnom + Delta;            %Here the uncertainty is additive.
%Gmult = Gnom*(1+Delta);        %Here the uncertainty is multiplicative.
step(Gadd,5), grid on, ylim([-5 5])
```

# Creation

# LTI Models

**Exercise 2.**

Let us consider a two-input, two-output system described in the state space by the equations

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where the matrices are next

```
A = [-1 0 5; 2 1 -4; -6 -3 -2]
B = [5 0; -4 1; 0 6]
C = [-1 0 -4; 2 3 6]
D = [3 -2; -4 1]
rank_CM = rank(ctrb(A, B))
rank_OM = rank(obsv(A, C))
```

This system is completely controllable and completely observable so that its state space model represents a minimal realization.

```
Gss = ss(A,B,C,D)
Gtf = tf(Gss)
```

```
p = pole(Gss)
z = tzero(Gss)
p = pole(Gtf)
z = tzero(Gtf)
G = ss(Gtf)
G = ss(Gtf,'min')
```

**Exercise 3**

Insert the next two-input two-output.

$$G = \begin{bmatrix} \frac{6}{(0.9s+1)(0.1s+1)} & \frac{-0.05}{0.1s+1} \\ \frac{0.07}{0.3s+1} & \frac{5}{(1.8s-1)(0.06s+1)} \end{bmatrix}$$

```
s = tf('s');
g11 = 6/((0.9*s + 1)*(0.1*s + 1));
g12 = -0.05/(0.1*s + 1);
g21 = 0.07/(0.3*s + 1);
g22 = 5/((1.8*s - 1)*(0.06*s+1));
```

The system is

```
G = [g11 g12; g21 g22]
```

## Students Intervention

What is the order of the system?

```
sigma(G,{10^(-2) 10^3})
title('Plant singular values')
grid
```

**Exercise 4.**

Consider the 2×2 transfer function matrix

$$G(s) = \begin{bmatrix} \frac{10(s+1)}{s^2+0.2s+100} & \frac{1}{s+1} \\ \frac{s+2}{s^2+0.1s+10} & \frac{5(s+1)}{(s+2)(s+3)} \end{bmatrix}$$

```
s = tf('s');
g11 = 10*(s + 1)/(s^2 + 0.2*s + 100);
g12 = 1/(s + 1);
g21 = (s + 2)/(s^2 + 0.1*s + 10);
g22 = 5*(s + 1)/((s + 2)*(s + 3));
G = [g11 g12 ;g21 g22];
```

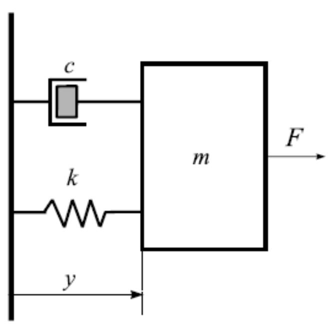
```
figure
sigma(G,{10^0 10^2})
grid
norm(G,'inf')
```

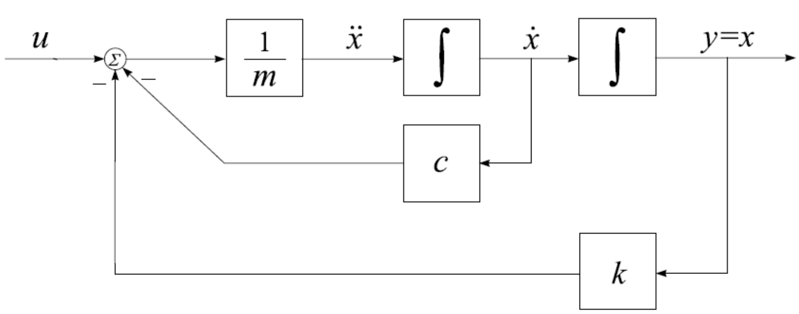
**Exercise 5.**

The mass-damper-system.

The dynamics of such a system can be described by the following second order differential equation, by Newton's Second Law

$$m\ddot{x} + c\dot{x} + kx = u$$




Denoting $x_1 = x, \quad x_2 = \frac{dx}{dt}$

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D = 0$$

The transfer function is given by

$$\frac{y(s)}{u(s)} = \frac{1}{ms^2 + cs + k}$$

In a realistic system, the three physical parameters *m*, *c*, and *k* are not known exactly. However, it can be assumed that their values are within certain, known intervals. That is,

$$m = \overline{m}(1 + p_m\delta_m), \qquad c = \overline{c}(1 + p_c\delta_c), \qquad k = \overline{k}(1 + p_k\delta_k)$$

```
% Parameters
m = ureal('m',3,'Percentage',[-40, 40])
c = ureal('c',1,'Range',[0.8, 1.2])
k = ureal('k',2,'PlusMinus',[-0.6, 0.6])

% Uncertain system
A = [ 0 1; -k/m -c/m];
B = [0 1/m]';
C = [1 0];
D = 0;
uss1 = ss(A,B,C,D)

%Uncertain transfer function
uss2 = tf(1,[m,c,k])
% Using the function feedback
s = tf('s');
g1 = (1/s)/m;
int2 = 1/s;
uss3 = feedback(int2*feedback(g1,c),k)


% Properties of uncertain model
get(uss1)
uss1.Uncertainty
uss1.NominalValue
SYS_1=usubs(uss1,'m',3,'c',1,'k',2)
```

```
% Bode plot of the uncertain plant
w = logspace(-1,1,100);
figure
bode(uss1,w), grid
title('Bode plot of uncertain system')
% Step responses of the uncertain plant
figure
step(uss1), grid
title('Step responses of uncertain system')
% Uncertain frequency response
w = logspace(-1,1,200);
freqs = ufrd(uss1,w);
figure
bode(freqs), grid
title('Uncertain frequency response')
% Uncertain frequency response
figure
```

```
frres = ufrd(uss2,w)
nyquist(frres), grid
title('Nyquist diagram of uncertain system')
% Step response of the mass-damper-spring system
% for a grid of 50 values of uncertain parameters
figure
step(gridureal(uss3,50)), grid
title('Step response for a grid of 50 values of uncertain parameters')
```